Labour Market Integration of Immigrants in Canada Post-Pandemic (2020–2024)

A Predictive and Industry-Level Analysis

(Proposal)

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Problem Statement & Context

The COVID-19 pandemic generated one of the most significant labour market disruptions in modern Canadian history, leading to widespread job losses, sectoral instability, and structural shifts in workforce demand. As Canada entered the post-pandemic recovery period (2020–2024), immigration levels rose in response to labour shortages and long-term economic planning. While aggregate employment indicators suggest substantial recovery, questions remain regarding the extent to which immigrants, particularly recent arrivals, have benefited from this recovery to the same extent as other population groups.

Labour market integration is a key indicator of economic stability, productivity, and long-term social inclusion. Drawing on my background in Economics, I approach this topic with an understanding of how macroeconomic shocks, labour demand fluctuations, and industry-specific recovery patterns shape employment outcomes. At the same time, my graduate training in Counselling Psychology highlights the behavioural and psychosocial dimensions of labour market integration, including stress, adaptation, motivation, and systemic barriers that influence employment trajectories. This interdisciplinary perspective strengthens the real-world relevance of the project by recognizing that labour market outcomes are shaped by both structural economic forces and human behavioural responses.

Despite extensive public discourse surrounding immigration and employment pressures, there remains a need for empirical, data-driven assessments of immigrant labour market outcomes during the post-pandemic recovery period. This project evaluates employment patterns among immigrants across Canada, examines industry-level concentration disparities, and applies predictive modeling techniques to identify structural factors associated with employment status. By combining descriptive labour market analysis with supervised machine-learning classification methods, this study provides an evidence-based assessment of immigrant labour market integration in Canada's post-pandemic economy.

Research Questions & Justification

This study addresses the following interconnected research questions:

RQ1:
How did employment rates among recent immigrants (0–5 years since arrival) change during the post-pandemic recovery period (2020–2024), and how do these trends compare to established immigrants and Canadian-born workers?

RQ2:
To what extent are recent immigrants concentrated in specific industries during the post-pandemic recovery, and how does this industry distribution differ from that of established immigrants and Canadian-born individuals?

RQ3:
Can supervised machine-learning models accurately predict employment status among working-age individuals during the recovery period, and which structural factors (education, industry, age, gender, immigrant status, year, and province) contribute most to these predictions?

Justification

These questions build on one another to create a clear analytical pathway. RQ1 establishes whether employment recovery patterns differed across immigrant groups. RQ2 explores whether these differences are linked to industry-level concentration or occupational segmentation. RQ3 then uses predictive modeling to quantify the relative importance of structural factors that shape employment outcomes. Together, these questions support a comprehensive, data-driven assessment of immigrant labour market integration in the post-pandemic period and align directly with the project's focus on predictive analytics.

Dataset Selection & Rationale

This project will use data from Statistics Canada's Labour Force Survey (LFS) for the years 2020–2024.

Source:

Statistics Canada. (2024). *Labour Force Survey (LFS) — Survey description (SDDS 3701)*. Government of Canada.
https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=3701

The LFS is a nationally representative monthly survey that provides detailed labour market information, including employment status, industry classification, education level, demographic characteristics, immigrant status, and province of residence. The dataset will include respondents across all Canadian provinces and territories.

Immigrant status will be categorized into three groups:

- Recent immigrants (0–5 years since arrival)
- Established immigrants (more than 5 years since arrival)
- Canadian-born individuals

"Province" will be retained as a predictor variable to account for potential regional variation; however, the primary focus of the analysis remains national-level patterns.

The LFS dataset is appropriate for this study because:

1. It directly measures employment outcomes central to the research questions.
2. It captures the post-pandemic recovery period (2020–2024).
3. It provides standardized industry and education classifications.
4. It is publicly accessible and methodologically documented.

Limitations include reliance on survey-based self-reporting and the inability to capture informal employment or credential recognition barriers. Nonetheless, the LFS remains the most comprehensive and authoritative source for labour market analysis in Canada.

Proposed Methodology & Tools

This study will employ a quantitative approach combining descriptive statistical analysis and supervised machine learning classification.

Data Preparation

- Include respondents across Canada (2020–2024).
- Categorize immigrant status (recent, established, Canadian-born).
- Encode categorical variables (industry, education level, gender, immigrant status, province).
- Include “year” as a predictor variable to capture recovery dynamics.
- Address missing data using appropriate preprocessing techniques.

Descriptive and Comparative Analysis

- Compare employment rates across immigrant groups.
- Visualize employment trends from 2020–2024.
- Analyze industry distribution across groups.
- Conduct chi-square tests to examine employment differences.
- Perform proportion tests to assess industry concentration disparities.

Predictive Modeling

Dependent Variable: employment status (employed vs unemployed).

Models:

- Logistic Regression (baseline model)
- Random Forest
- Gradient Boosting

Evaluation Metrics:

- Accuracy
- Precision

- Recall
- F1-score
- ROC-AUC
- Cross-validation

Feature importance analysis will identify the structural factors most strongly associated with employment outcomes. If feasible, SHAP values may be applied to enhance model interpretability.

This combined statistical and predictive framework ensures alignment between research questions, dataset characteristics, and methodological rigor.

References

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